FILE PERMISSIONS

File permissions in Linux define who can access a file or directory and what actions they are allowed to perform. They are an important part of Linux security and access control.

Each file or directory has **three types of users**:

- **User (u)** – the user who created the file
- **Group (g)** – users belonging to the same group
- **Others (o)** – all other users

Each user type can have **three permissions**:

- **Read (r)** – allows viewing file content
- **Write (w)** – allows modifying file content
- **Execute (x)** – allows running a file or accessing a directory

Permissions are commonly viewed using the ls -l command and modified using the **chmod** command.

Linux supports both **symbolic mode** (r, w, x) and **numeric mode** (4, 2, 1) for setting permissions.

HANDS-ON FOR FILE PERMISSIONS

Image 1: Create Directory and File & View Permissions

- Checked current directory using pwd
- Created a directory using mkdir
- Listed files using ls
- Used ls -l (starts with d means directory)
- Moved into directory using cd
- Created a file using touch
- Verified file and permissions using ls and ls -l

```
        _/ _/
      _/m/'
[ec2-user@ip-172-31-45-198 ~]$ pwd
/home/ec2-user
[ec2-user@ip-172-31-45-198 ~]$ mkdir folder1
[ec2-user@ip-172-31-45-198 ~]$ ls
folder1
[ec2-user@ip-172-31-45-198 ~]$ ls -l
total 0
drwxr-xr-x. 2 ec2-user ec2-user 6 Mar  3 01:36 folder1
[ec2-user@ip-172-31-45-198 ~]$ cd folder1
[ec2-user@ip-172-31-45-198 folder1]$ touch file2
[ec2-user@ip-172-31-45-198 folder1]$ ls
file2
[ec2-user@ip-172-31-45-198 folder1]$ touch file file3
[ec2-user@ip-172-31-45-198 folder1]$ ls
file  file2  file3
[ec2-user@ip-172-31-45-198 folder1]$ ls -l
total 0
-rw-r--r--. 1 ec2-user ec2-user 0 Mar  3 01:38 file
-rw-r--r--. 1 ec2-user ec2-user 0 Mar  3 01:38 file2
-rw-r--r--. 1 ec2-user ec2-user 0 Mar  3 01:38 file3
```

Image 2: Changing File Permissions Using Symbolic Mode

- Created additional files using touch
- Added execute permission to **user (owner)** using chmod u+x file
- Added write permission to **group** using chmod g+w file2
- Added execute permission to **others** using chmod o+x file3
- Verified permission changes using ls -l

```
-rw-r--r--. 1 ec2-user ec2-user 0 Mar  3 01:38 file3
[ec2-user@ip-172-31-45-198 folder1]$ chmod u+x file
[ec2-user@ip-172-31-45-198 folder1]$ ls -l file
-rwxr--r--. 1 ec2-user ec2-user 0 Mar  3 01:38 file
[ec2-user@ip-172-31-45-198 folder1]$ chmod g+w file2
[ec2-user@ip-172-31-45-198 folder1]$ ls -l file2
-rw-rw-r--. 1 ec2-user ec2-user 0 Mar  3 01:38 file2
[ec2-user@ip-172-31-45-198 folder1]$ chmod o+w file3
[ec2-user@ip-172-31-45-198 folder1]$ ls -l file3
-rw-r--rw-. 1 ec2-user ec2-user 0 Mar  3 01:38 file3
[ec2-user@ip-172-31-45-198 folder1]$ ls -l
total 0
-rwxr--r--. 1 ec2-user ec2-user 0 Mar  3 01:38 file
-rw-rw-r--. 1 ec2-user ec2-user 0 Mar  3 01:38 file2
-rw-r--rw-. 1 ec2-user ec2-user 0 Mar  3 01:38 file3
[ec2-user@ip-172-31-45-198 folder1]$
```

Image 3: Changing File Permissions Using Numeric Mode

- Set permission **700** using chmod 700 file (full access to owner only)
- Set permission **471** using chmod 471 file2
- Set permission **007** using chmod 007 file3 (full access to others only)
- Verified permission changes using ls -l

```
[ec2-user@ip-172-31-45-198 folder1]$
[ec2-user@ip-172-31-45-198 folder1]$
[ec2-user@ip-172-31-45-198 folder1]$ chmod 700 file
[ec2-user@ip-172-31-45-198 folder1]$ chmod 471 file2
[ec2-user@ip-172-31-45-198 folder1]$ chmod 007 file3
[ec2-user@ip-172-31-45-198 folder1]$ ls -l
total 0
-rwx------. 1 ec2-user ec2-user 0 Mar  3 01:38 file
-r--rwx--x. 1 ec2-user ec2-user 0 Mar  3 01:38 file2
-------rwx. 1 ec2-user ec2-user 0 Mar  3 01:38 file3
[ec2-user@ip-172-31-45-198 folder1]$
[ec2-user@ip-172-31-45-198 folder1]$
[ec2-user@ip-172-31-45-198 folder1]$
```

Image 4: Changing File Owner and Group

- Changed file owner to **root** using

  chown root file
- Changed file group to **root** using

  chown :root file2
- Changed both owner and group using

  chown root:root file3
- Verified ownership changes using ls -l

```
[ec2-user@ip-172-31-45-198 folder1]$
[ec2-user@ip-172-31-45-198 folder1]$ sudo su
[root@ip-172-31-45-198 folder1]# chown root file2
[root@ip-172-31-45-198 folder1]# ls -l
total 0
-rwx------. 1 ec2-user ec2-user 0 Mar  3 01:38 file
-r--rwx--x. 1 root     ec2-user 0 Mar  3 01:38 file2
-------rwx. 1 ec2-user ec2-user 0 Mar  3 01:38 file3
[root@ip-172-31-45-198 folder1]#  chown :root file2
[root@ip-172-31-45-198 folder1]# ls -l
total 0
-rwx------. 1 ec2-user ec2-user 0 Mar  3 01:38 file
-r--rwx--x. 1 root     root     0 Mar  3 01:38 file2
-------rwx. 1 ec2-user ec2-user 0 Mar  3 01:38 file3
[root@ip-172-31-45-198 folder1]# chown root:root file
[root@ip-172-31-45-198 folder1]# ls -l
total 0
-rwx------. 1 root     root     0 Mar  3 01:38 file
-r--rwx--x. 1 root     root     0 Mar  3 01:38 file2
-------rwx. 1 ec2-user ec2-user 0 Mar  3 01:38 file3
[root@ip-172-31-45-198 folder1]# touch file4
[root@ip-172-31-45-198 folder1]# ls -l
total 0
-rwx------. 1 root     root     0 Mar  3 01:38 file
-r--rwx--x. 1 root     root     0 Mar  3 01:38 file2
-------rwx. 1 ec2-user ec2-user 0 Mar  3 01:38 file3
-rw-r--r--. 1 root     root     0 Mar  3 01:58 file4
[root@ip-172-31-45-198 folder1]# history
    1  chown root file2
    2  ls -l
    3   chown :root file2
    4  ls -l
    5  chown root:root file
    6  ls -l
    7  touch file4
    8  ls -l
    9  history
[root@ip-172-31-45-198 folder1]#
```

Image 5: Adding a New User in Linux

- Switched to root user using sudo su
- Created a new user named **jai** using adduser jai
- Navigated to home directory using cd /home
- Verified user creation using ls

```
[ec2-user@ip-172-31-45-198 folder1]$
[ec2-user@ip-172-31-45-198 folder1]$  sudo su
[root@ip-172-31-45-198 folder1]# adduser jai
[root@ip-172-31-45-198 folder1]# cd /home
[root@ip-172-31-45-198 home]# ls
ec2-user  jai
[root@ip-172-31-45-198 home]# cd ..
```

Image 6: Viewing Users Using /etc/passwd

- Displayed user account details using

  cat /etc/passwd
- Verified the newly created user entry in the system

```
lp:x:4:7:lp:/var/spool/lpd:/sbin/nologin
sync:x:5:0:sync:/sbin:/bin/sync
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown
halt:x:7:0:halt:/sbin:/sbin/halt
mail:x:8:12:mail:/var/spool/mail:/sbin/nologin
operator:x:11:0:operator:/root:/sbin/nologin
games:x:12:100:games:/usr/games:/sbin/nologin
ftp:x:14:50:FTP User:/var/ftp:/sbin/nologin
nobody:x:65534:65534:Kernel Overflow User:/:/sbin/nologin
dbus:x:81:81:System message bus:/:/sbin/nologin
systemd-network:x:192:192:systemd Network Management:/:/usr/sbin/nologin
systemd-oom:x:998:998:systemd Userspace OOM Killer:/:/usr/sbin/nologin
systemd-resolve:x:193:193:systemd Resolver:/:/usr/sbin/nologin
rpc:x:32:32:Rpcbind Daemon:/var/lib/rpcbind:/sbin/nologin
libstoragemgmt:x:997:997:daemon account for libstoragemgmt:/:/usr/sbin/nologin
systemd-coredump:x:996:996:systemd Core Dumper:/:/usr/sbin/nologin
systemd-timesync:x:995:995:systemd Time Synchronization:/:/usr/sbin/nologin
sshd:x:74:74:Privilege-separated SSH:/usr/share/empty.sshd:/sbin/nologin
chrony:x:994:994:chrony system user:/var/lib/chrony:/sbin/nologin
ec2-instance-connect:x:993:993::/home/ec2-instance-connect:/sbin/nologin
stapunpriv:x:159:159:systemtap unprivileged user:/var/lib/stapunpriv:/sbin/nologin
rpcuser:x:29:29:RPC Service User:/var/lib/nfs:/sbin/nologin
tcpdump:x:72:72::/:/sbin/nologin
ec2-user:x:1000:1000:EC2 Default User:/home/ec2-user:/bin/bash
jai:x:1001:1001::/home/jai:/bin/bash
jon:x:1002:1002::/home/jon:/bin/bash
[root@ip-172-31-45-198 folder1]#
```

Image 7: Viewing a Specific User

- Displayed user details using

  cat /etc/passwd | grep jai
- Filtered and verified information for the specific user

```
[root@ip-172-31-45-198 folder1]# cat /etc/passwd |grep jon
jon:x:1002:1002::/home/jon:/bin/bash
[root@ip-172-31-45-198 folder1]#
```

Image 8: Adding a New Group

- Created a new group named **newgp** using

  groupadd newgp
- Verified group creation using

  cat /etc/group

```
[root@ip-172-31-45-198 folder1]#
[root@ip-172-31-45-198 folder1]# groupadd newgp
[root@ip-172-31-45-198 folder1]# cat /etc/group
```

```
tcpdump:x:72:
screen:x:84:
ec2-user:x:1000:
jai:x:1001:
jon:x:1002:
newgp:x:1003:
[root@ip-172-31-45-198 folder1]#
```

Image 9: Viewing a Specific Group

- Displayed group details using

  cat /etc/group | grep newgp
- Verified the newly created group in the system

```
[root@ip-172-31-45-198 folder1]#
[root@ip-172-31-45-198 folder1]# cat /etc/group | grep newgp
newgp:x:1003:
[root@ip-172-31-45-198 folder1]#
```

Image 10: Setting Password for a User

- Entered password for user jai
- Re-entered the new password for confirmation
- Password updated successfully (authentication tokens updated)

```
[root@ip-172-31-45-198 folder1]#
[root@ip-172-31-45-198 folder1]# passwd jon
Changing password for user jon.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
[root@ip-172-31-45-198 folder1]#
```

Image 11: Verifying User Password

- Verified password entry for user jai using

  cat /etc/shadow | grep jai
- Confirmed that the password is set and encrypted

```
[root@ip-172-31-45-198 folder1]#
[root@ip-172-31-45-198 folder1]# cat /etc/shadow |grep jon
jon:$6$B0JieAuUC.UoMlDQ$zXtR76Pq6fuSPSVG8/y3p6cJMtBGifJNpmjyczjLvrNYsnCfKWaDq62lYr.Ax6LhdnnRrnJ7YyR4AD./5ToaP0:20515:0:99999:7:::
[root@ip-172-31-45-198 folder1]#
[root@ip-172-31-45-198 folder1]#
[root@ip-172-31-45-198 folder1]#
```

Image 12: User Without Password

- Created another user without setting a password
- Checked password status using

  cat /etc/shadow | grep <username>
- Verified that no password is set for the user

```
[root@ip-172-31-45-198 folder1]#
[root@ip-172-31-45-198 folder1]# adduser surya
[root@ip-172-31-45-198 folder1]# cat /etc/shadow |grep surya
surya:!!:20515:0:99999:7:::
[root@ip-172-31-45-198 folder1]#
```